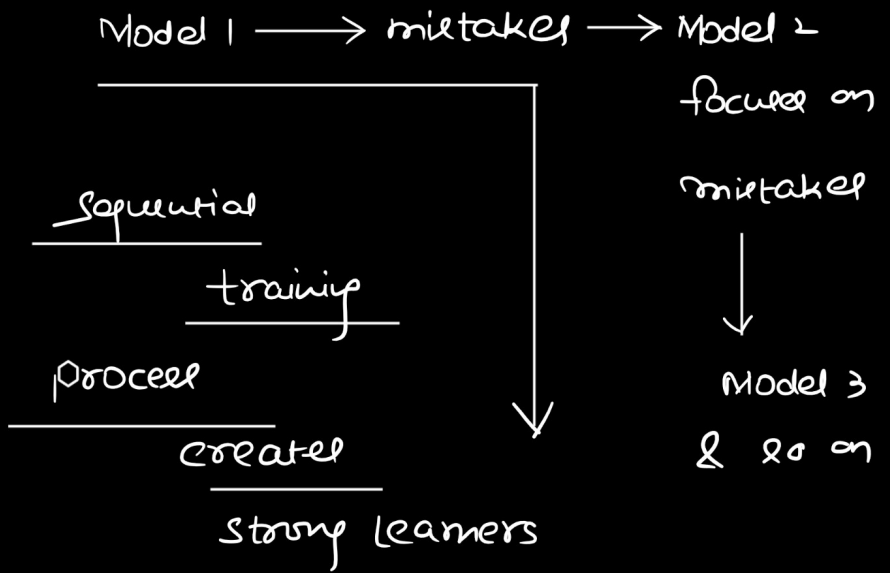
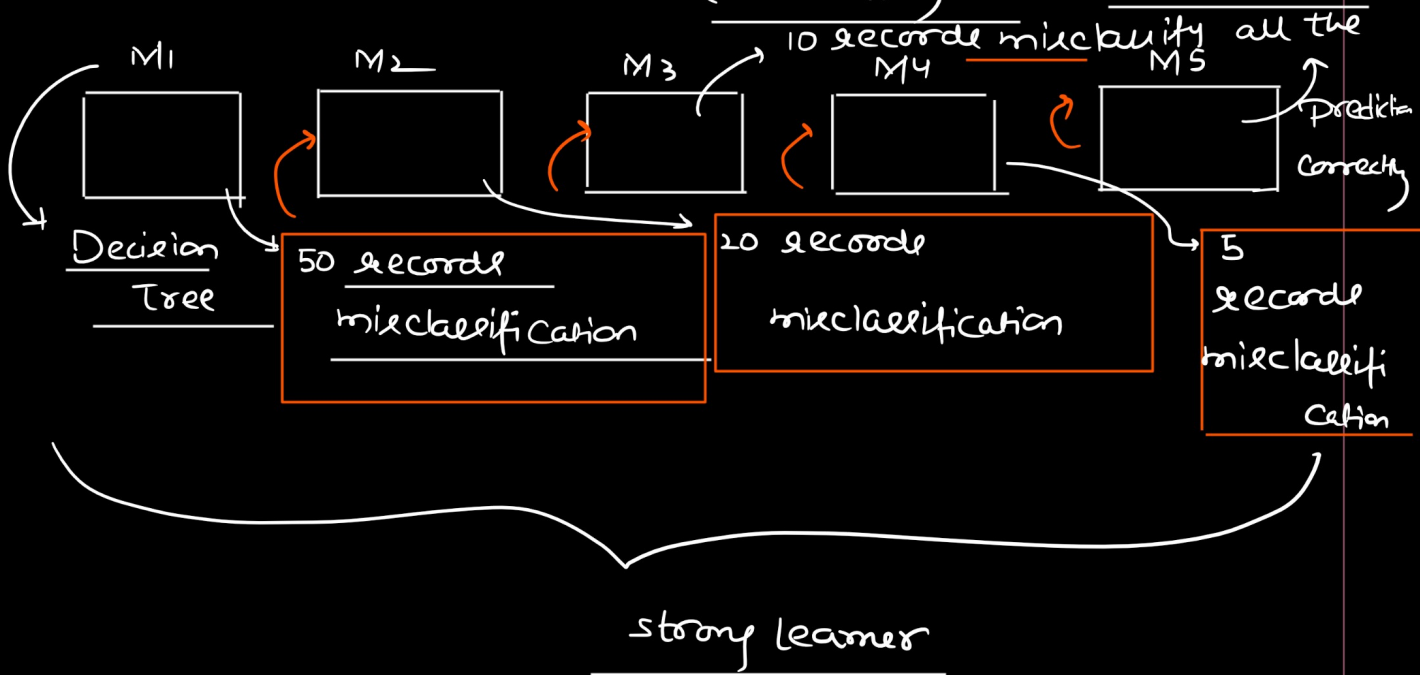


Boosting Algorithms (Classification & Regression)

→ reduce the biasness (training set error)

(underfitted)



- Advanced
- ① Adaboost ✓
 - ② Gradient Boosting ✓
 - ③ Xgboost (Extreme Gradient Boosting) ✓
- } sklearn separate library

LightGBM

catboost

error = 0.2 + 0.2 = 0.4

Adaboost (Adaptive Boosting)

random

$M_1 = 1/h = 1/5$

f_1	f_2	y	$\hat{y}$	weight	updated wt- M_1	norm wt- M_1	range- M_1
3	7	1	1	0.2	0.16	0.1666	0 - 0.166
2	9	0	1	0.2	0.24	0.25	0.166 - 0.416
1	4	1	0	0.2	0.24	0.25	0.416 - 0.666
9	8	0	0	0.2	0.16	0.1666	0.666 - 0.832
3	7	0	0	0.2	0.16	0.1666	0.832 - 1
					0.96		

$M_1 \rightarrow$ DT (max-depth = 1)
Weak learner

$M_2 \rightarrow$ DT

more focus towards misclassified points

less focus towards correctly classified points

overall prediction =

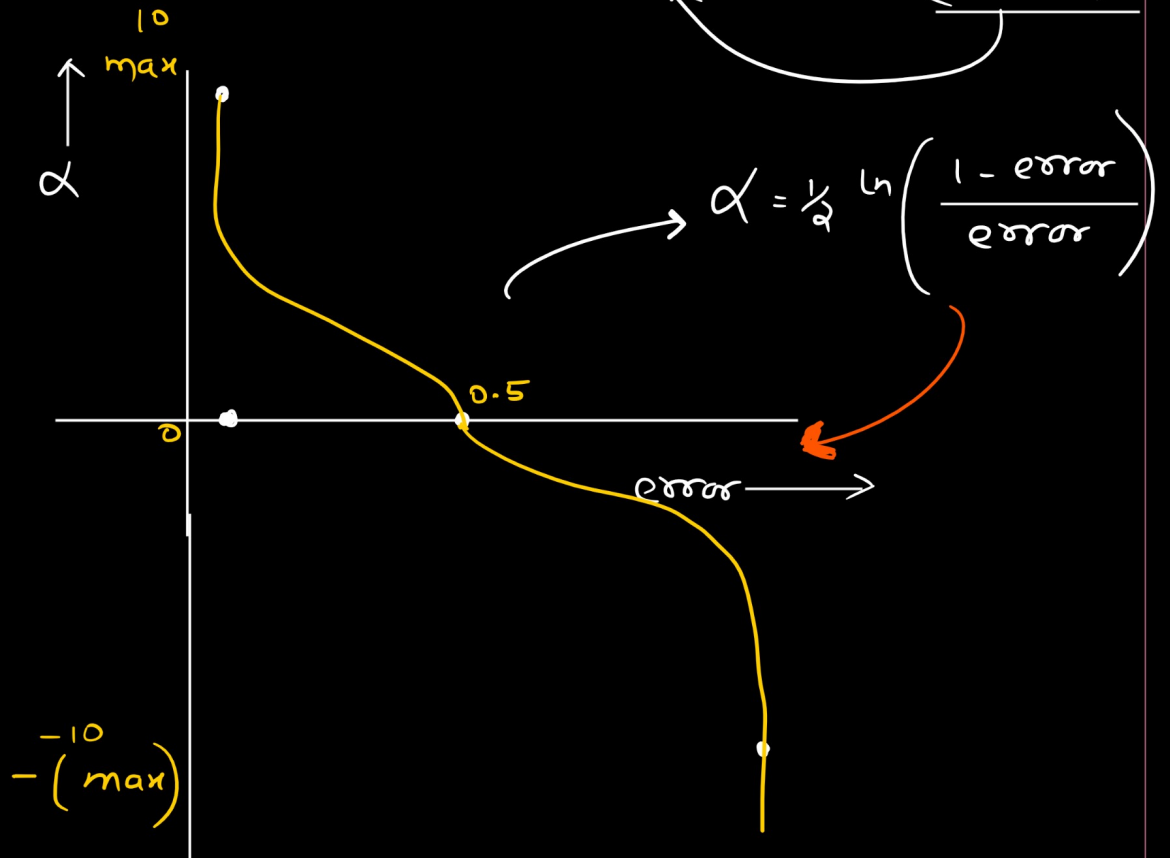
$\alpha_1 m_1 + \alpha_2 m_2 + \alpha_3 m_3$

$\alpha \rightarrow$ contribution of each model towards making prediction
error

error $\rightarrow 0\%$ $\rightarrow \alpha(\text{max})$
 $\hookrightarrow$ model's say

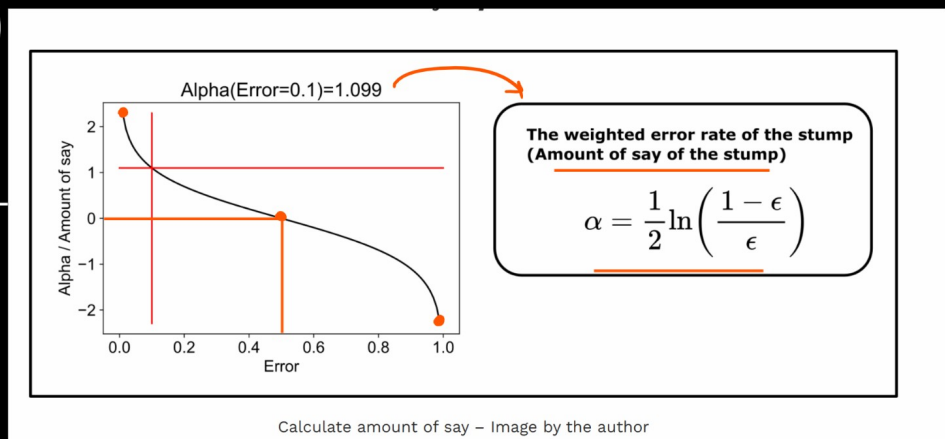
error $\rightarrow 50\%$ $\rightarrow \alpha = 0$

error $\rightarrow 100\%$ $\rightarrow \alpha(\text{reverse})$



$$\alpha_1 = \frac{1}{2} \ln \left(\frac{1 - 0.4}{0.4} \right)$$

$$\Rightarrow 0.20$$



for any misclassified points

$\hookrightarrow$ new weights = curr_wt $\times e^{\alpha_1}$

$$\Rightarrow 0.2 \times e^{0.20} \Rightarrow 0.24$$

for any classified points

$$\hookrightarrow \text{new weights} = \text{curr-}\omega_t * e^{-\alpha_1}$$

$$= 0.2 * e^{-0.20}$$

$$\Rightarrow \underline{0.16}$$

M_2

randomly

Decision stump

f_1	f_2	y
2	9	0
2	9	0
2	9	0
1	4	1
1	4	1

Gradient Boosting

f_1	f_2	y	$\hat{y}_1$ baseline	R_1	m_2 $\hat{y}_2$	pred_2	R_2	$\hat{y}_3$
Intelligence score	Exam score	Package	$M_1 \text{ pred}$					M_3
90	8	<u>3</u>	4.8	-1.8	-1.8	<u>4.62</u>	-1.62	-1.62
100	7	4	4.8	-0.8	-0.8	<u>4.72</u>	-0.72	-0.72
110	6	<u>8</u>	4.8	3.2	3.2	<u>5.12</u>	2.88	2.88
120	9	<u>6</u>	4.8	1.2	1.2	<u>4.92</u>	1.08	1.08
80	5	3	4.8	-1.8	-1.8	<u>4.62</u>	-1.62	-1.62

pred_3

4.458

4.648

5.408 ✓

6.108 ✓

4.458

Error / Residual $\rightarrow (y - \hat{y})$

$M_2 \rightarrow$ Decision Tree Regressor

